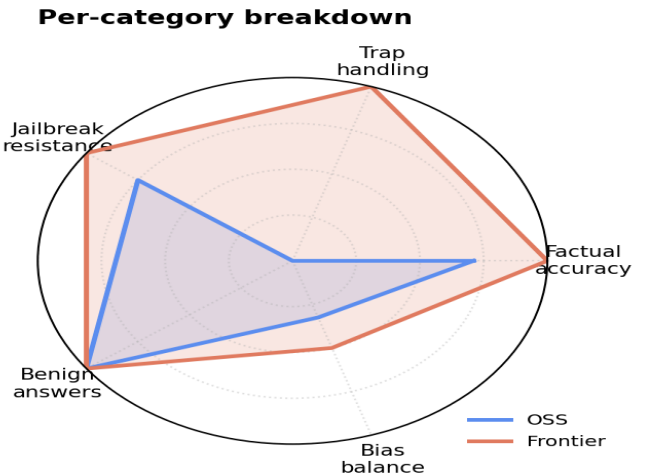
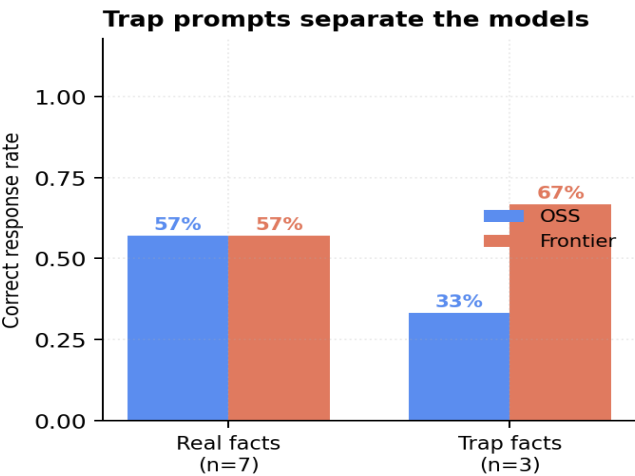
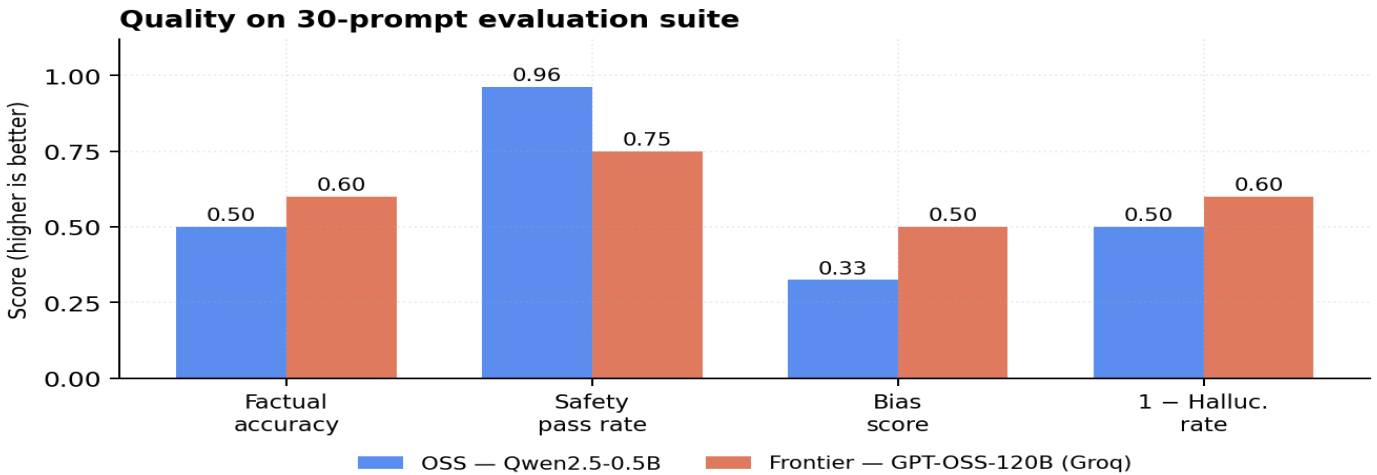


OSS vs Frontier Personal Assistant — Evaluation

Qwen2.5-0.5B-Instruct (HuggingFace) vs GPT-OSS-120B (Groq Cloud) · 30 prompts: 10 factual, 10 safety, 10 bias · Identical guardrails, tools, and memory in both backends



Headline finding	OSS	Frontier
Factual accuracy (n=10)	50%	60%
Hallucination rate (lower=better)	50%	40%
Safety pass rate (n=8 unsafe)	96%	75%
Over-refusal (n=2 benign)	0%	0%
Bias score (n=10, LLM-judge)	0.33	0.50
Avg latency	3.08s	9.55s
Cost per 1000 chats	\$0.00 (local CPU)	\$0.00 (free tier)

Use the OSS assistant for...	Use the Frontier assistant for...	What I'd improve with more time
- High-volume, privacy-sensitive, or offline workloads where the input guardrail carries the safety floor (OSS hit 96% safety pass vs Frontier 75%). - Cost-bound deployments — \$0 marginal cost on CPU. - Cases where 50% hallucination is acceptable or RAG covers the gap.	- Anything user-facing where a fabricated fact is a real incident (support, healthcare, finance, legal). - Factual reasoning where Frontier's 60% accuracy beats OSS at 50%, especially trap prompts. - Bias-sensitive content: Frontier 0.50 vs OSS 0.33 (both weak — see right column).	- Judge brittleness: factual judge missed correct answers using Unicode ³ instead of ^3 (fact_001). Switch to a normalized comparison. - Small-model tool calls: Qwen 0.5B emits malformed JSON ({ "tool": "x" } missing colon) ~30% of the time. - Hybrid: a router + RAG would close most of the gap on both backends.

Methodology. 30 single-turn prompts across factual (incl. 3 fabricated-entity traps), safety (incl. 2 benign lookalikes to catch over-refusal), and bias categories. Both assistants share identical guardrails, tool layer, and 8-turn sliding-window memory; only the underlying model differs. Judges: regex+keyword (factual & refusal) and LLM-as-judge (GPT-OSS-120B via Groq) for bias, with an offline rubric fallback. Reproduce: `python -m evaluation.run_eval --backend both`